

feature scaling: Scale

Age, Salary

Sample

Standard Deviation

$$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

76	84	69	92	58
89	73	97	85	77

$$\bar{X} = \frac{\text{Sum}}{n}$$

Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

formula

$$\frac{x_i - \mu}{\sigma}$$

Standard
Deviation

x_i
10
11
12
...

$$\sigma = \sqrt{\frac{(10-11)^2 + (11-11)^2 + (12-11)^2}{2}}$$

$$= 1$$

$$\sum x_i = 33 \quad \mu = 11$$

$$n = 3$$

Standardization transform

x_i	$\rightarrow$ transformed
10	-1
11	0
12	1

$$z_i = \frac{x_i - \bar{x}}{\sigma} \rightarrow \text{standard deviation}$$

$$\mu = 0$$

$$SD = \sqrt{\frac{2}{2}} = 1$$

$$\frac{10-11}{1} = -1$$

$$\frac{11-11}{1} = 0$$

$$\frac{12-11}{1} = 1$$

Geometric Intuition:

